Lab 2 - Theory reading plan

Estimation of Motion Variables of the Parrot AR.Drone (MEAer, UAVs, 2026/27)

Goal: arrive at Sections 3, 4 and 5 of the Lab 2 guide knowing every concept the questions rely on. Tick each box when done.

How to read this sheet

SLIDES A lecture deck PDF in `lectures-22-23/`. The page number is the **PDF page** of that deck (page 1 is the cover).

BOOK `reference-book/uavbook.pdf` (Beard and McLain). Two page numbers: the one **printed** on the page, and the **PDF** page (PDF = printed + 8).

NOTES A written summary PDF or `.tex` in `lectures-22-23/notes/` or `labs-25-26/notes/`. Not slides, not the book.

PAPER The Batista, Silvestre and Oliveira article (Section 5 of the guide).

The files used

Short name	File (relative to the repository root)	Pages
Course Ch5-6 deck	`lectures-22-23/22_23_UAVs_Ch5-6_Sensors_and_State_Estimation_1.pdf`	28
Beard Sensors deck	`lectures-22-23/22_23_UAVs_Ch5_Sensors_Beard_McLain.pdf`	29
Beard Estimation deck	`lectures-22-23/22_23_UAVs_Ch6_State_Estimation_Beard_McLain.pdf`	47
Course Ch4 deck	`lectures-22-23/22_23_UAVs_Ch4_Modern_Control_Design.pdf`	43
Book	`reference-book/uavbook.pdf`	401
Sensors/estimation notes	`lectures-22-23/notes/03-sensors-estimation.pdf`	-
Lab 2 notes (25/26)	`labs-25-26/notes/02-lab2.pdf` (and `.tex`)	-
Paper	`labs-25-26/Partial attitude and rate gyro bias estimation ... .pdf`	10

There are no recorded video lectures in the repository.

Map: guide questions → areas

Guide question	Area to read first
3.1, 3.2, 3.3 (sensor mean and covariance, experiments A, B, C)	Area 1
3.4 (raw pitch and roll from the accelerometer)	Area 2
4.1, 4.2, 4.3 (steady-state Kalman filter, transfer functions, Q and R)	Areas 3, 4, 5
4.4 (filter with gyro bias state)	Areas 5, 6, 7
4.5 (complementary filters)	Area 6
4.6, 4.7 (experiments A to E, roll version)	Areas 1 to 6
5 (partial attitude and gyro bias, the paper)	Area 7

Area 0 - Prerequisites (skim, about 30 min)

☐	Type	Source and topic	Pages
☐	SLIDES	`Ch1_Rigid_Body` deck: attitude, Euler angles, rotation matrices	whole deck (skim)
☐	SLIDES	`Ch2_Quadrotor_Modeling` deck: kinematics, relation between body rates (p, q, r) and Euler-angle rates	whole deck (skim)
☐	SLIDES	Course Ch4 deck: linear state-space models, observability	3 to 30
☐	SLIDES	Course Ch4 deck: Luenberger observer example	41 to 43

Can I do this without looking?

☐ Write the small-angle relation between the gyro measurements and $\dot{\phi}, \dot{\theta}$, and say when it breaks down.

Area 1 - Sensor models: accelerometer, rate gyro, bias, noise (guide 3.1 to 3.3)

☐	Type	Source and topic	Pages
☒	SLIDES	Beard Sensors deck: accelerometer principle, measured acceleration is specific force (5 to 6), accelerometer model (9 to 10), rate gyro principle and gyro model with bias (11 to 13)	3 to 13
☒	BOOK	Chapter 7 "Sensors for MAVs", sections 7.1 Accelerometers and 7.2 Rate Gyros	printed 121 to 127; PDF 129 to 135
☒	SLIDES	Course Ch5-6 deck: AR.Drone sensor suite	3 to 4
☒	SLIDES	Course Ch5-6 deck: accelerometer and rate gyro principle	6 to 9
☐	NOTES	`03-sensors-estimation.pdf`: accelerometer and rate gyro sections (1.2, 1.3 and 2.1, 2.2)	2 to 3 and 5

Can I do this without looking?

- ☐ Why does a resting accelerometer read about $-g$ on the z axis, and why is that not a bias?
- ☐ What does the accelerometer measure during a manoeuvre, and why does that corrupt the inclinometer?
- ☐ Explain bias, noise standard deviation and noise covariance, and what a diagonal covariance implies.

Area 2 - Raw roll and pitch (inclinometer) (guide 3.4)

☐	Type	Source and topic	Pages
☐	SLIDES	Beard Estimation deck: inverting the sensor measurement model; roll and pitch inversion under steady level flight is on 9 and 10	6 to 12
☐	BOOK	Chapter 8 "State Estimation", sections 8.2 Low-pass Filters and 8.3 State Estimation by Inverting the Sensor Model	printed 148 to 153; PDF 156 to 161

Can I do this without looking?

- ☐ Derive θ from a_x and ϕ from a_y, a_z under the steady-flight assumption.
- ☐ Say when this estimate fails (manoeuvres, vibration), and which of our experiments shows it.

Area 3 - Kalman filter and observers (guide 4.1 to 4.3)

☐	Type	Source and topic	Pages
☐	SLIDES	Beard Estimation deck: dynamic observer (13 to 15), quadratic forms and Gaussian (17 to 18), continuous-discrete Kalman filter derivation and covariance update (19 to 29)	13 to 29
☐	BOOK	Chapter 8, sections 8.4 Dynamic-observer Theory and 8.5 Derivation of the Continuous-discrete Kalman Filter	printed 154 to 160; PDF 162 to 168
☐	SLIDES	Course Ch5-6 deck: Kalman filter	12 to 13
☐	SLIDES	Course Ch5-6 deck: duality between LQR and Kalman filter	14
☐	NOTES	`03-sensors-estimation.pdf`: Kalman filter (1.5, 3.3, 3.4) and practical tuning advice (3.9)	3, 7 to 8 and 10

Can I do this without looking?

- ☐ What does the steady-state Kalman filter assume that a time-varying one does not?
- ☐ How do Q and R change the gain, and what does a very large R/Q ratio do?
- ☐ In MATLAB, what do `kalman()` and `estim()` take and return?

Area 4 - LQR and its dual

(guide 4.1; also Lab 3)

☐	Type	Source and topic	Pages
☐	SLIDES	Course Ch4 deck: linear state-space optimal control	26
☐	SLIDES	Course Ch4 deck: modern control design and LQR design example	31 to 40
☐	BOOK	Chapter 6 “Autopilot Design”, section 6.3 LQR Control	printed 115 to 117; PDF 123 to 125

Can I do this without looking?

- ☐ State the LQR/Kalman duality: $K = R^{-1}B^{T}P$ versus $L = \Sigma C^{T}N^{-1}$, and the two Riccati equations.

Area 5 - Attitude filter and tuning Q, R

(guide 4.1, 4.3, 4.7)

☐	Type	Source and topic	Pages
☐	SLIDES	Course Ch5-6 deck: worked roll-angle example. This is the template for the lab	15 to 23
☐	SLIDES	Beard Estimation deck: attitude estimation with EKF	33 to 37
☐	SLIDES	Beard Estimation deck: suggestions for tuning your EKF	47
☐	BOOK	Chapter 8, section 8.8 Attitude Estimation	printed 164 to 166; PDF 172 to 174

Can I do this without looking?

- ☐ Name the state, the input and the measurement in the roll example, and which sensor plays which role.
- ☐ For constant gains, write the two transfer functions from inclinometer and from gyro to the estimate.
- ☐ Which of Q, R would I raise for experiment B (vibration), and why?

Area 6 - Complementary filters

(guide 4.5)

☐	Type	Source and topic	Pages
☐	SLIDES	Course Ch5-6 deck: complementary filters	24 to 26
☐	SLIDES	Course Ch5-6 deck: complementary filters and Kalman filters	27 to 28
☐	NOTES	`03-sensors-estimation.pdf`: section “Complementary filters” (1.7) and key equations	4 and 11

The book has no dedicated section on this (checked in its table of contents).

Can I do this without looking?

- ☐ State what “complementary” means ($G_L + G_H = 1$) and where it fails.
- ☐ Explain why a complementary filter is a Kalman filter with fixed gain.

Area 7 - Gyro bias and partial attitude estimation

(guide 4.4 and Section 5)

☐	Type	Source and topic	Pages
☐	PAPER	Batista, Silvestre, Oliveira, *Int. J. Control* 84(5), 2011: problem statement (Sec. 2), observability analysis (Sec. 3, Theorem 3.3), filter design (Sec. 4), simulation results (Sec. 5)	journal pp. 895 to 903 (10 PDF pages)
☐	NOTES	`labs-25-26/notes/02-lab2.tex`: walkthrough of the paper, including the observability theorem (Theorem 3.3 on p. 4, filter on p. 5)	3 to 5

The book has no dedicated section on bias estimation (checked in its table of contents).

Can I do this without looking?

- [ ] Why can a motionless drone not determine the full gyro bias (Theorem 3.3)?
- [ ] Relate that to our data: experiments A, B, C (parked or hovering) versus E (mission).
- [ ] The filter state is the vector observation and the bias, not roll and pitch: how do I recover roll and pitch?

Lab 3 pointers (not part of this lab)

☐	Type	Source and topic	Pages
☐	BOOK	Chapter 10, section 10.1 Straight-line Path Following	printed 198 to 203; PDF 206 to 211
☐	SLIDES	`Ch7_Intro_Nonlinear_Control` and `Ch7_Intro_Nonlinear_Control_Part_II` decks: Lyapunov theory and LaSalle (Part I), Lyapunov analysis and adaptive control (Part II)	Part I 17 to 31; Part II 14 to 18

Suggested order and time

The times are rough estimates, not measurements.

☐	Step	Content	Time
☐	1	Area 0 and Area 1	about 2 h
☐	2	Area 2 and Area 3	about 2 h
☐	3	Area 4 and Area 5	about 3 h
☐	4	Area 6	about 1 h
☐	5	Area 7 (the paper)	about 2 h
☐	6	Redo every “Can I do this without looking?” box	about 1 h

Minimal path if short on time

- ☐ 1. SLIDES Beard Sensors deck, pages 3 to 13.
- ☐ 2. SLIDES Beard Estimation deck, pages 6 to 12, then 13 to 29.
- ☐ 3. SLIDES Course Ch5-6 deck, pages 15 to 23, then 24 to 28.
- ☐ 4. PAPER The Batista article.

Page numbers were checked against the PDFs on 2026-10-18 (notes and Lab 3 pages added on 2026-10-29): slide titles page by page, and book section headings located by text search. Pages of the notes PDFs are PDF pages.